

Michele Tosi

Quantitative Researcher – Behavioral Data Analyst

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Skills

- **Experimentation & Causal Inference:** Quantitative analytics, A/B style testing, experimental design, sampling, randomization, quasi-experimental designs, mixed-effects models, measurement.
- **UX & Behavioral Research:** Expertise in survey-based research, sampling theory, survey design, scale validation, measurement invariance. Experience in usability testing, moderated user sessions, behavioral and audience segmentation (clustering, latent class), AI-assisted analysis, advanced quantitative data integration.
- **Programming & Machine Learning:** Python (pandas, scikit-learn, statsmodels, numpy), R, Excel, SPSS, basic SQL, Qualtrics; NLP, classification, visualization tools (matplotlib, seaborn, ggplot), LLMs (word embeddings, machine psychology, benchmarking), explainable AI, sentiment analysis.
- **Mixed Methods:** Qualitative analytics, semi-structured interviews, data insights, AI prompting, AI analysis.

Work Experience

Quantitative Researcher, Italian Institute of Technology, Italy Oct 2021 – Apr 2026

- Leading quantitative research projects (A/B-style) combining behavioral data analysis, surveys, and biometric sensor data across 180+ participant sessions. *One study showed a repeatable 10% gain in performance and traced it to a single cause, giving the team clear evidence on which method to keep.*
- Iterated protocols from structured user debriefs, improving session tolerability and data quality.
- Built reproducible Python/R pipelines (mixed-effects modeling, time series, feature engineering).

Visiting Scientist – Quantitative Researcher, University of Oxford, UK Apr 2025 – Oct 2025

- Modeled physiological, behavioral, imaging, and clinical data to predict functional outcomes at the individual level; competitively awarded fellowship.

Statistical Consultant, Freelancer Sep 2023 – Sep 2024

- Undergraduate consultant on experimental design and statistics; standardized Python/R workflows.

Survey Specialist, University of Padova & University of Verona, Italy Oct 2018 – Oct 2021

- Delivered survey infrastructure and psychometric validation end-to-end; performed behavioral segmentation via clustering and dimensionality reduction on multilingual samples to drive actionable behavioral insights.

Selected Projects

- **Survey Optimization at Scale** — Validated a 42-item instrument, $N = 39,775$ ($\alpha = 0.88\text{--}0.96$); Random Forest feature importance for item selection, reducing respondent burden while preserving predictive signal.
- **Cross-Demographic Survey Validation** — Established measurement invariance across sex and country ($N = 1,420$, multilingual), ensuring segment comparisons reflect genuine behavioral differences, *so demographic groups could be compared with confidence rather than survey noise.*
- **LLMs Classification with Integrity Diagnostics** — NLP text classification pipeline benchmarking network vs. lexical features; flagged performance inflation from LLM stylistic homogeneity. *Caught that a top-scoring model was misleading, avoiding a decision built on inflated numbers.* Delivered an LLM evaluation framework with NLP-based sentiment classification and explainable cognitive feature extraction.
- **Behavioral & Attitude Tracking Framework** — Managed the quantitative pipeline ($N = 1,420$) and complex statistical analysis for a cross-national project, isolating personality-driven variance in user decision-making and risk perception.

Education

- **PhD in Brain and Cognitive Sciences**, University of Trento (CIMEC), Italy
- **MSc in Data Science**, University of Trento, Italy
- **MSc in Cognitive Psychology**, University of Padova, Italy
- **BSc in Industrial/Organizational Psychology**, University of Verona, Italy

Additional Training & Languages

Professional training: Scuola Superiore Sant'Anna, Pisa — Data Science-driven Innovation in Business & Society (2026); University of Genoa — Machine Learning Crash Course (2025); Radboud University, Nijmegen — Biometric Signal Processing & Behavioral Analytics. Grants: IIT Mobility, Erasmus+, 4-year PhD Fellowship. **Languages:** Italian (native), English (C1).